Project Report

FMCG Retail Store Segmentation Using Clustering

1. Abstract

Retail stores differ in sales scale, demand variability, growth patterns, and product-category preferences. Treating every store identically may limit the effectiveness of assortment planning and store-level business decisions. This project applies unsupervised machine learning to transaction data to identify groups of FMCG retail stores with similar characteristics.

The dataset consists of 117 transaction CSV files containing approximately 31 million transaction records. After preprocessing and aggregation, 761 unique stores were analyzed. Store-level features were derived from weekly spend, sales variability, log spend change, and product-category mix. K-Means and Agglomerative Clustering were compared using internal clustering metrics, cluster sizes, interpretability, and sensitivity analysis.

The selected K-Means two-cluster solution identifies 139 smaller-spend stores and 622 larger-spend stores. The segments differ in typical weekly spend, absolute and relative sales variability, and store-format composition. These findings provide a foundation for targeted assortment reviews and controlled business experiments. The results are descriptive and do not establish causal relationships or directly measure profitability.

2. Introduction

Fast-Moving Consumer Goods (FMCG) retailers operate stores with different sales volumes, formats, regions, and product preferences. A uniform strategy may not adequately reflect these differences. Store segmentation groups similar stores so that business teams can investigate suitable actions for each group.

Transaction data provides information about purchasing activity, spend patterns, and product-category preferences. By transforming transaction-level records into store-level features, clustering algorithms can identify groups with similar observed characteristics.

This project develops a workflow covering data preprocessing, exploratory analysis, feature engineering, clustering, model evaluation, segment profiling, and business recommendations. The objective is to produce interpretable store segments and a reproducible analytical process.

3. Problem Statement

To segment FMCG retail stores using transaction data and unsupervised machine learning, evaluate the quality and stability of alternative clustering solutions, interpret the resulting store groups, and propose evidence-based business actions.

4. Objectives

The main objectives are:

1. Inspect and preprocess the transaction files.
2. Perform exploratory data analysis and data-quality checks.
3. Build meaningful store-level features from transaction records.
4. Compare K-Means and Agglomerative Clustering across different cluster counts.
5. Select and profile an interpretable clustering solution.
6. Examine sensitivity to random initialization and feature selection.
7. Generate store assignments, segment profiles, and a recommendation matrix.
8. Identify data limitations that should be addressed before making operational decisions.

5. Dataset Description

The input consists of 117 transaction CSV files containing approximately 31 million transaction records.

Dataset characteristic	**Reported value**
Transaction files	117
Approximate transaction records	31 million
Unique stores analyzed	761
Distinct products in processed data	4,997
Product-category groups	250
Store-week records	88,551
Store-category records	173,765
Weekly observations	117

The weekly aggregated dataset was previously inspected as covering 10 April 2006 to 6 July 2008.The raw transaction date range should be verified independently before a definitive date range is stated for the complete input dataset.

6. Data Preprocessing and Quality Assessment

The preprocessing workflow standardizes inconsistent column names and processes large transaction files in chunks to reduce memory requirements. Transactions are aggregated into store-week-product, store-week, and store-category datasets. These aggregated datasets provide the basis for store-level feature engineering.

Data-quality checks examine missing values, duplicate store-week records, store coverage, feature distributions, and unusually large growth values.

The initial store-feature dataset contains 761 stores. The reported checks found no duplicate store identifiers or duplicate store-week records in the corresponding aggregated datasets. The store-week dataset contains 88,551 records across 761 stores and 117 weeks.

A key issue concerns stores with no recorded spend in the first 13-week comparison period. Ten stores were identified with zero starting-period spend and positive spend in the final comparison period. Conventional percentage growth is undefined when starting spend is zero, so these stores should not be assigned an ordinary percentage growth rate.

The clustering workflow treats the log spend change of zero-starting-spend stores as missing and uses median imputation for missing numerical feature values. This prevents undefined growth values from being treated as valid observed growth measurements.

7. Exploratory Data Analysis

The initial exploratory analysis identified substantial variation in store performance.

Feature	Reported value
Mean total units per store	Approximately 61,509
Mean total spend per store	Approximately 78,428
Mean average weekly spend across stores	Approximately 674.08
Mean weekly spend volatility across stores	Approximately 153.69
Mean active weeks	Approximately 116.36 out of 117

The reported store-format distribution is:

Store format	Number of stores
LS	330
MS	236
SS	167
XLS	28
Total	761

The stores span 12 regions.

Several spend-related variables are highly correlated. Total spend and average weekly spend had a reported correlation of approximately 0.998, while total units and total spend had a reported correlation of approximately 0.995. Such correlations suggest that including every raw scale feature could introduce redundant information into the clustering process.

The modeling workflow therefore uses selected and transformed numerical features rather than including all raw measures indiscriminately.

The analysis also identified unusually large growth values. These observations were reviewed because small starting-period spend can inflate relative growth calculations and because zero starting spend makes conventional percentage growth undefined.

8. Feature Engineering

Store-level features were constructed to represent sales scale, variability, changes in spend, and product-category preferences.

8.1 Average weekly spend

Average weekly spend represents the typical sales scale of a store over its observed history. It helps distinguish stores with different levels of recorded spending.

8.2 Log-transformed average spend

The feature log_avg_weekly_spend is calculated using:

log1p(max(avg_weekly_spend, 0))

This transformation reduces the influence of highly skewed spend values.

8.3 Relative spend volatility

Relative spend volatility is calculated as:

weekly_spend_volatility / avg_weekly_spend

A zero average spend results in an undefined ratio, which is handled as a missing value during preprocessing. Relative volatility pro

vides a scale-adjusted measure of variation in weekly spend.

8.4 Log spend change

Log spend change is calculated as:

log1p(last_13_week_spend) - log1p(first_13_week_spend)

It captures the change between the two comparison periods on a logarithmic scale. It is not an ordinary percentage growth rate.

For stores with zero starting-period spend, the log spend change is treated as missing for clustering because the starting-period comparison is not suitable for interpreting conventional growth.

8.5 Product-category mix

Product-category mix represents the proportion of a store's spend associated with selected product categories. The 15 categories with the highest total spend across stores were selected for the clustering feature matrix.

These features allow the model to consider differences in product-category preferences in addition to sales behavior.

8.6 Historical activity and contextual features

Active weeks, observed weeks, store format, store region, the starting-period spend flag, and a short-history flag are retained for quality checks and interpretation.

Not all these fields are included as numerical inputs to the clustering model. In the current modeling script, the numerical feature matrix uses log-transformed average spend, relative spend volatility, log spend change, and the selected category-share features.

8.7 Imputation and scaling

Infinite values are replaced with missing values. Missing numerical values are imputed using the median of each feature, and the resulting feature matrix is standardized using StandardScaler.

This process places numerical features on comparable scales before clustering.

9. Clustering Methodology

Two unsupervised clustering algorithms were evaluated.

9.1 K-Means

K-Means partitions stores into a specified number of clusters by minimizing within-cluster squared distances to cluster centroids. It provides a straightforward approach to identifying groups of stores with similar numerical feature profiles.

The implementation uses random_state=42 and n_init=20 to make initialization behavior reproducible under the same data and preprocessing configuration.

9.2 Agglomerative Clustering

Agglomerative Clustering builds a hierarchy of groups by progressively merging observations. The implementation uses Ward linkage, which merges groups according to increases in within-cluster variance.

9.3 Cluster-count evaluation

Cluster counts from two through eight were evaluated for both algorithms.

The evaluation considered:

- Silhouette score.
- Davies–Bouldin score.
- Calinski–Harabasz score.
- Smallest and largest cluster sizes.
- Interpretability of the resulting groups.
- Sensitivity to random initialization and feature selection.

The silhouette score measures how well observations fit their assigned clusters compared with neighboring clusters; higher values generally indicate better separation and cohesion.

The Davies–Bouldin score measures similarity between clusters, with lower values generally preferred.

The Calinski–Harabasz score compares between-cluster dispersion with within-cluster dispersion, with higher values generally preferred.

No single metric establishes that a clustering solution is universally best. The scores must be interpreted alongside cluster sizes, feature distributions, and business usefulness.

10. Model Comparison and Selection

The reported initial model comparison produced the following results.

Algorithm	Number of clusters	Silhouette	Davies–Bouldin	Calinski–Harabasz
K-Means	2	0.3248	1.8196	155.6441
K-Means	3	0.3164	1.6044	114.4322
Agglomerative	3	0.2940	1.7446	107.2781
Agglomerative	2	0.2910	2.0147	144.2133

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K-Means with two clusters achieved the highest silhouette score among the listed candidates and the highest Calinski–Harabasz score in this comparison. The three-cluster K-Means solution had a lower reported Davies–Bouldin score, indicating that the two-cluster model did not outperform every alternative on every metric.

The two-cluster solution was selected as the primary segmentation because it provided the strongest silhouette result in the reported comparison and produced two groups that could be interpreted in terms of spend scale and variability.

The three-cluster solution was retained as a comparison to acknowledge that alternative segmentations may capture additional differences between stores.

11. Final Segment Profiles

Characteristic	Smaller-Spend Stores	Larger-Spend Stores
Number of stores	139	622
Share of stores	18.27%	81.73%
Median store-level average weekly spend	142.17	751.51
Mean store-level average weekly spend	171.89	786.31
Median weekly spend volatility	45.06	160.45
Median relative spend volatility	0.33	0.21
Median active weeks	117	117
Stores with zero starting-period spend	0	10

The selected K-Means two-cluster solution assigns all 761 stores to two segments.

11.1 Smaller-Spend Stores

The smaller-spend segment contains 139 stores, or approximately 18.27% of all stores. Its median store-level average weekly spend is substantially lower than that of the larger-spend segment.

The segment also has lower absolute weekly spend volatility but higher relative spend volatility. This indicates that weekly spending varies more in relation to typical spend, even though the absolute variation is smaller.

Potential business actions include reviewing category assortment, investigating potential product gaps, and testing targeted changes in a controlled group of comparable stores.

11.2 Larger-Spend Stores

The larger-spend segment contains 622 stores, or approximately 81.73% of all stores. Its mean and median average weekly spend are higher than those of the smaller-spend segment.

This group also exhibits higher absolute weekly spend volatility but lower relative spend volatility. The distinction matters because large stores can experience larger absolute changes simply because their sales scale is higher.

Potential actions include reviewing category-specific assortment, examining replenishment requirements, and evaluating operational service levels using additional inventory and availability data.

11.3 Store-format composition

The reported format analysis indicates that approximately 66.9% of smaller-spend stores are SS format, while approximately 52.4% of larger-spend stores are LS format.

This association suggests that store format should be considered when interpreting the segments. It does not demonstrate that store format causes a particular spend level.

The labels “Smaller-Spend Stores” and “Larger-Spend Stores” are descriptive labels for this dataset and should not be treated as universal classifications.

12. Stability and Sensitivity Analysis

The project evaluated clustering across multiple random seeds and with and without the store-scale feature.

The reported results are:

Configuration	Number of clusters	Mean silhouette	Mean adjusted Rand index
With store scale	2	0.3245	0.9984
With store scale	3	0.3167	0.9973
Without store scale	2	0.3122	1.0000
Without store scale	3	0.3131	0.9916

The adjusted Rand index (ARI) measures agreement between clustering assignments. Values closer to 1 indicate stronger agreement, while a value of 1 indicates identical partitions.

The reported results suggest that the clustering was highly consistent across the tested random seeds. Removing the store-scale feature slightly reduced the mean silhouette score for the two-cluster configuration, while assignment agreement remained high.

In a separate reproducibility check, a fresh K-Means two-cluster run was compared with the saved kmeans_2_assignments.csv file. All 761 stores were matched, and the adjusted Rand index was 1.0. This confirms that the fresh run reproduced the saved partition under the tested feature preparation and model configuration.

The sensitivity analysis evaluates initialization and a particular feature-selection choice. It does not establish robustness under dataset resampling, new time periods, or changes in the underlying retail environment.

13. Business Recommendations

The segments provide a basis for proposing targeted actions. These recommendations should be treated as hypotheses to test rather than proven outcomes.

13.1 Recommendations for smaller-spend stores

- Review the assortment of major product categories and identify potential gaps.
- Compare category mix with similar stores in the same format and region.
- Pilot targeted assortment changes or promotions in a small group of comparable stores.
- Track weekly spend, unit sales, category mix, and variability before and after the pilot.
- Compare outcomes with a suitable control group before expanding the intervention.

13.2 Recommendations for larger-spend stores

- Review high-contribution product categories and investigate opportunities for category-specific assortment planning.
- Examine replenishment requirements using additional inventory and availability data.
- Test assortment improvements while monitoring weekly spend, units sold, and operational service levels.
- Review performance across store formats and regions rather than applying one uniform strategy.
- Avoid assuming that higher spend necessarily implies higher profit.

13.3 Recommendations for both segments

- Use store format and region to refine the interpretation of each cluster.
- Define measurable success criteria before launching each business experiment.
- Use comparable stores or control groups to assess the impact of interventions.
- Reassess segment assignments when more recent transaction data becomes available.
- Monitor whether stores' sales patterns change over time rather than assuming that their original segment remains permanent.

14. Limitations

14.1 Data and business-feature limitations

The verified fields described in this report include transaction spend, quantities, product categories, store format, and store region. Price sensitivity, basket type, and dominant shopping mission are not included as established data fields because their availability has not been verified.

The following business features have not been verified in the available data:

- **Promotions and discounts:** Explicit promotion identifiers and reliable promotion-effectiveness measures have not been verified.
- **Pricing:** Transaction spend is available, but comparable unit selling prices and discount amounts have not been validated.
- **Inventory and stockouts:** Inventory-on-hand and stockout records have not been verified.
- **Distribution and product availability:** Explicit store-product distribution and availability records have not been verified.
- **Profitability:** Product cost and margin fields have not been verified.

Consequently, sales spikes cannot automatically be attributed to promotions, low sales cannot automatically be interpreted as low demand, and the resulting segments should be interpreted as sales-behavior segments rather than profitability segments.

14.2 Growth outliers and interpretation

Store spend was compared between the first and last 13-week periods. Previously identified examples include STORE01856, whose spend increased from 222.26 to 9,097.53, and STORE00962, whose spend increased from 252.23 to 7,647.69.

The saved feature analysis reported a maximum log-spend-change value of approximately 39.93 for STORE01856. These figures describe different measures: the log-spend-change feature is not an ordinary percentage growth rate.

Extreme percentage increases can arise when the starting-period spend is small. For stores with zero starting-period spend, conventional percentage growth is undefined. The analysis identified ten such stores with positive final-period spend.

The observed patterns may reflect changes in store activity, incomplete observations, or other factors. Transaction data alone cannot establish the cause of each extreme value. These cases should therefore be flagged for investigation rather than automatically treated as data errors or evidence of successful business performance.

Growth estimates also depend on the selected 13-week comparison windows and the availability of observations. Store opening dates, store activity records, and operational data would help improve interpretation.

14.3 Modeling limitations

Clustering is an unsupervised method, so there are no predefined ground-truth segment labels against which to measure classification accuracy.

The model-selection metrics evaluate different aspects of cluster structure and do not guarantee business usefulness. The two-cluster model performs best on the reported silhouette comparison, but another configuration may reveal additional meaningful differences.

The selected features and scaling choices also influence the resulting clusters. Although initialization and feature-selection sensitivity were examined, further validation using resampling, later time periods, and additional operational variables would strengthen confidence in the segmentation.

14.4 Business interpretation limitations

The current analysis is descriptive. It does not establish that a particular store characteristic causes higher or lower spend, nor does it prove that a recommended intervention will improve business performance.

Promotion response, inventory constraints, product availability, and profit margins should be incorporated into future analysis where reliable data becomes available.

15. Conclusion

This project developed a workflow for FMCG retail store segmentation using transaction data and unsupervised machine learning. The analysis covered 761 stores and compared K-Means and Agglomerative Clustering across multiple cluster counts.

K-Means with two clusters was selected as the primary solution based on its reported silhouette score and the interpretability of the resulting groups. The selected segmentation distinguishes 139 smaller-spend stores from 622 larger-spend stores, with differences in typical spend, relative variability, and store-format composition.

A separate reproducibility check achieved an adjusted Rand index of 1.0 when comparing a fresh two-cluster K-Means run with the saved assignments. Additional work is needed to ensure that the complete pipeline automatically generates all required assignment files and reports.

The final segment assignments and recommendation matrix provide a foundation for targeted business experiments. Future work should incorporate verified margin, promotion, inventory, and pricing data and evaluate whether segment-specific actions improve measurable business outcomes.

16. Project Deliverables

The project is intended to produce the following deliverables:

- Processed transaction aggregates and enriched store features.
- Data-quality and exploratory data analysis reports.
- Clustering model comparison results.
- K-Means two-cluster and three-cluster store assignments.
- Final store-to-segment assignments.
- Segment profiles by store format, region, and product category.
- Stability and sensitivity analysis.
- Growth-feature audit and outlier-review reports.
- Segment recommendation matrix.
- Project documentation, including setup instructions, execution steps, methodology, results, and limitations.

17. Reproducibility and Execution

The project uses Python with pandas, NumPy, and scikit-learn for data preparation and clustering. Matplotlib and Seaborn are available for visualization.

The workflow includes the following stages:

- Process raw transaction files and create aggregated datasets.
- Build store-level features.
- Generate enriched features and review growth-related data quality.
- Run the clustering comparison.
- Generate cluster assignments and profiles.
- Run stability and sensitivity analysis.

- Create final segment assignments, profiles, and recommendations.
- Validate the saved results and confirm that the documentation matches the generated outputs.

The exact commands and their execution order should be documented in the project README. Reproducibility requires using the same input data, feature preparation, model parameters, and software environment. Any random seed, feature-selection choice, or preprocessing change should be recorded so that its impact can be evaluated.

18. Future Enhancements

Future work could improve the project in several ways:

- Incorporate verified unit prices, promotions, discount amounts, and product margins.
- Include inventory, stockout, and product-availability information.
- Compare segments across later time periods to evaluate whether assignments remain stable.
- Evaluate alternative feature sets and clustering methods using consistent preprocessing.
- Use store-format and region-specific analysis to identify more actionable patterns.
- Run controlled business experiments to determine whether segment-specific assortment or replenishment changes improve measurable outcomes.
- Automate the complete pipeline, including assignment generation, validation, report creation, and reproducibility checks.

These enhancements would help bridge the gap between descriptive clustering and operational decision-making.

19. Final Summary

The project demonstrates how transaction-derived store features can be used to create interpretable FMCG store segments. The selected two-cluster solution identifies groups with different sales scales and relative spend variability. The results support further investigation into assortment planning and store-specific business actions, while acknowledging that profitability and causal impact cannot be inferred from the current analysis alone.

The project should be considered complete for submission only after all reported statistics have been checked against the saved outputs, duplicate documentation has been removed, and the documented workflow has been tested from the processed inputs through the final deliverables.